PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer | NLP and Document AI

petra.zlataric@example.com | +385 55 555 0606 | github.com/pzn-ml-example

PROFESSIONAL SUMMARY

Machine learning engineer with more than five years of experience in NLP and document classification, spanning research, model development and production deployment. Production experience with transformer fine-tuning, FastAPI model serving, Docker, Azure and MLflow. Previous customer complaint classification experience in banking. PhD in Computer Science in progress, focused on named entity recognition in Croatian legal texts.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o.
03/2023 - Present

- Built and deployed an incoming-document classifier for public administration: 94% accuracy and 0.91 macro F1 across 23 classes, processing approximately 15,000 documents per day after OCR.
- Fine-tuned XLM-R for document classification and BERTić for named entity recognition in Croatian administrative documents.
- Served models through FastAPI in Docker containers on Azure App Service; introduced MLflow experiment tracking and a model registry, using Azure ML in the deployment workflow.
- Built an annotated dataset of 4,000 documents with annotation guidelines and two annotators, achieving inter-annotator kappa of 0.87.
- Implemented production monitoring with weekly reports on text length, vocabulary and predicted class distributions, with alerts when PSI exceeded 0.2. Detected an OCR change that reduced accuracy by 9 percentage points; the issue was resolved within a week.
- Mentor two student interns.

Data Scientist | Baranjska kreditna banka d.d.
09/2021 - 02/2023

- Developed customer complaint topic classification using TF-IDF and logistic regression, followed by fastText, covering 11 topics for weekly customer service management reporting.
- Built customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Produced SQL reports from an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, Sveučilište Drava
10/2020 - 08/2021

- Created a corpus and annotations of Croatian legal texts and wrote Python scripts to support the research.

SELECTED PERSONAL PROJECT

Retrieval-Augmented Question Answering Demo | 2025

- Built a weekend prototype for question answering over Croatian laws using sentence embeddings, FAISS and an open-weight LLM accessed through an API.
- Personal RAG exploration only; not deployed to production and assessed through manual checks only.

EDUCATION

PhD in Computer Science | Sveučilište Drava
2022 - Present | Expected completion: 2027
Research: Named entity recognition in Croatian legal texts.

Master's Degree in Computer Science (Mag. ing. comp.) | Sveučilište Drava
2020

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate | 2024

TECHNICAL SKILLS

Programming and data: Python, SQL, pandas, NumPy, Oracle SQL, PostgreSQL
Machine learning and NLP: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, TF-IDF, sentence-transformers, FAISS
Production ML and MLOps: Azure ML, Azure App Service, Azure Functions, MLflow, FastAPI, Docker, Evidently, Git, Linux
Annotation: Label Studio, annotation guidelines, inter-annotator agreement

KNOWLEDGE SHARING

- Guest lecturer at an NLP summer school, delivering a 90-minute session in 2025.
- Co-authored a 2024 regional workshop paper on named entity recognition for Croatian administrative documents.

LANGUAGES

Croatian: Native | English: C1 | German: B2
Daily English-language meetings with a client in Slovenia.